

Program : Diploma in Civil and Rural Engineering	
Course Code : 4368	Course Title: Irrigation & Drainage Engineering Lab
Semester : 4	Credits: 1.5
Course Category: Common Course	
Periods per week: 3(L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To enable the students to familiarize the irrigation structures
- To enable the students to familiarize the drainage systems

Course Prerequisites:

Topic	Course code	Course name	Semester
Engineering drawing	1005	Engineering graphics	1
		Building and sanitation Engineering	

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Draw the longitudinal and cross section of the Irrigation structures	12	Applying
CO2	Draw the culverts and Canal head works	6	Applying
CO3	Draw the building drainage system	12	Applying
CO4	Open ended Project	9	Applying
	Lab test	6	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Draw the longitudinal and cross section of the Irrigation structures		
M1.01	Draw the cross section of masonry dam showing component parts	3	Applying
M1.02	Draw the longitudinal section and Plan of a canal drop showing component parts	3	Applying
M1.03	Draw the longitudinal section and Plan of a Tank sluice with tower head	3	Applying
M1.04	Draw the longitudinal section and plan of a surplus escape/weir, showing component parts	3	Applying
CO2	Draw the culverts and Canal head works		
M2.01	Draw detailed sectional plan, sectional elevation, and side elevation of a slab culvert with return wing wall.	2	Applying
M2.02	Draw detailed sectional plan, sectional elevation and side elevation of a slab culvert with splayed wing wall.	2	Applying
M2.03	Layout plan of Canal Head works	2	Applying
	Lab Test-1	3	
CO3	Draw the building drainage system		

M3.01	Draw the Layout of house drainage arrangement	3	Applying
M3.02	Draw the Layout of Sanitary pipe work above ground-One pipe system, Two pipe system and single stack system	3	Applying
M3.03	Draw the layout of a domestic sewage system	3	Applying
M3.04	Draw the plan and show the different components of a septic tank and dispersion trench	3	Applying
CO4	Measurement of in situ hydraulic conductivity		
M4.01	Open ended Project	9	Applying
	Lab Test-2	3	

Text / Reference:

T/R	Book Title/Author
R1	Irrigation Engineering and Hydraulic Structures by Santosh Kumar Garg
R2	Irrigation Water Resources and Water Power Engineering by P. N. Modi
R3	Irrigation and Water Power Engineering eBook : Das, Madan Mohanmimi Das Saikia
R4	Irrigation and Water Power Engineering. B. C. Punmia
R5	A course in Civil Engineering Drawing.V.B.Sikka by S.K. Kataria & Sons

Online resources:

Sl.No.	Website Link
1	https://aktu.ac.in/pdf/syllabus/Syllabus1819/agricultureeng3rd/FIFTH%20SEMESTER.pdf
2	https://dbskkv.org/pdf/Faculty/Engineering/Irrigation%20and%20Drainage%20Engineering1.pdf